

Incident Ticket #3: Local Network Printer Inaccessible

Status: Resolved

Grade / Evaluation: A (Efficient diagnostic flow; identified missing VLAN database entry preventing access port operation)

Ticket Overview

- **Ticket ID:** TCK-003
- **Source:** Reception PC1 & PC2 | **VLAN:** 10
- **Destination:** Office Printer | **Interface:** Switch A Fa0/1 | **Target VLAN:** 40 (Printers)
- **Symptom:** Reception staff unable to reach or send print jobs to the local network printer.
- **Expected Behavior:** Unicast routing and layer 2 forwarding to the printer on VLAN 40.
- **Actual Behavior:** Destination host unreachable / 100% packet loss.

Root Cause Analysis

Interface Fa0/1 on Switch A was assigned to VLAN 40, but **VLAN 40 did not exist in Switch A's local VLAN database (vlan.dat)**. Because VLAN 40 was not active in the local VLAN database, Switch A could not properly forward traffic for VLAN 40 on that access port.

Resolution Sequence

[Reception PCs] ---> [Switch A] -X (VLAN 40 Missing) X-> [Printer]

|
(Provisioned VLAN 40)

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[Reception PCs] ---> [Switch A] ---> [Printer Interface Fa0/1]

1. **Physical Verification:** Checked physical cabling; link lights active.
2. **Database Provisioning:** Added VLAN 40 to Switch A's database and applied it to access interface Fa0/1:

```
Switch-A(config)# vlan 40
Switch-A(config-vlan)# name Printers
Switch-A(config-vlan)# exit
Switch-A(config)# interface FastEthernet0/1
Switch-A(config-if)# switchport mode access
Switch-A(config-if)# switchport access vlan 40
Switch-A(config-if)# no shutdown
```

3. **Configuration Persistence:**

```
Switch-A# copy running-config startup-config
```

Verification Proof

```
C:\> ping 192.168.40.10
```

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=64

Reply from 192.168.40.10: bytes=32 time=2ms TTL=64

Reply from 192.168.40.10: bytes=32 time<1ms TTL=64

Reply from 192.168.40.10: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.40.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Key Technical Takeaways

- An interface set to **switchport access vlan X** will not forward traffic unless VLAN X is explicitly declared in **show vlan brief**.
- Depending on IOS behavior, an undefined access VLAN may be created automatically or appear inactive. Best practice is to explicitly create and name the VLAN before assigning ports.